ICON + HAM-lite: Interactive aerosols in a global storm-resolving model

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- Simple, interactive aerosol module, HAM-lite *[Weiss et al., GMD (2025)]* coupled to global storm resolving model ICON *[Segura et al., GMD (2025)]*

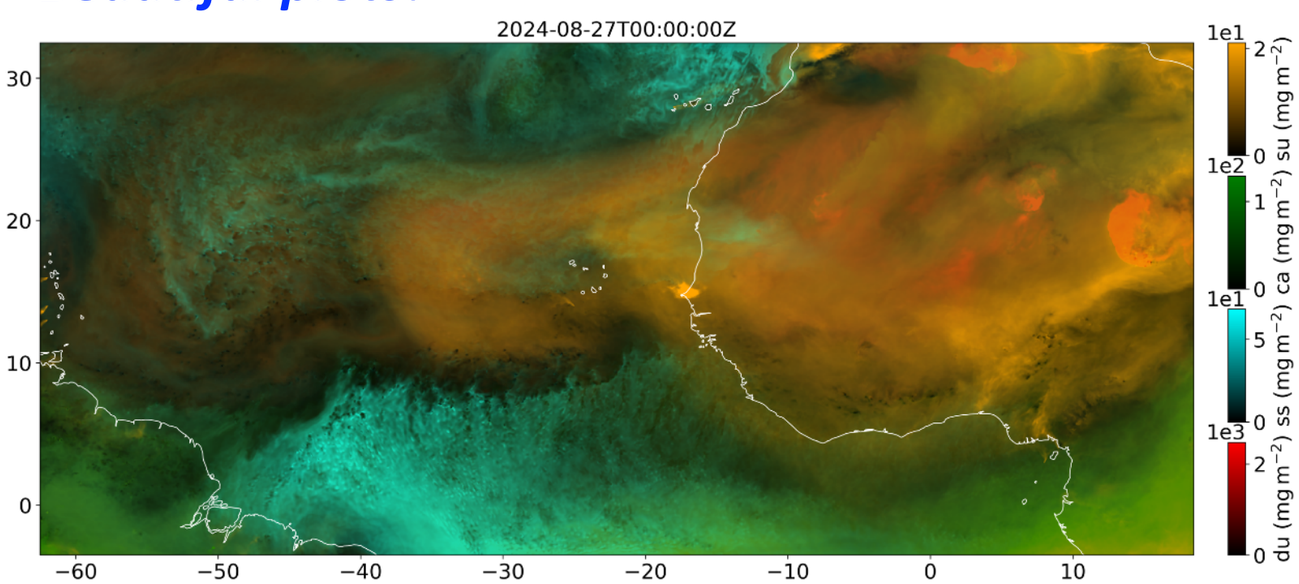
ICON-MPIM	HAM-lite
Cloud microphysics	Wet deposition and Activation
Radiation	Optical properties
Turbulence	Dry deposition and Emission
	Sedimentation

Parameterized processes in ICON and HAM-lite

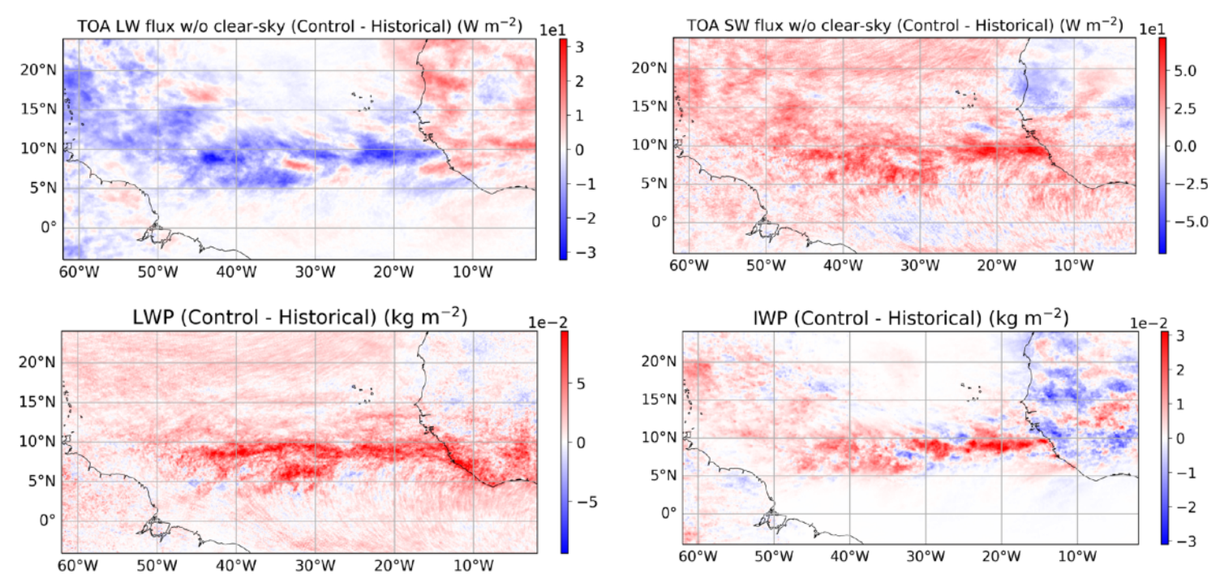
MODES	Hydrophilic
AITKEN (0.005 μm < r < 0.05 μm)	
ACCUMULATION (0.05 μm < r < 0.5 μm)	N_{BB}, N_{AV}
COARSE (0.5 μm < r)	N_{SS}, N_{DU}

Aerosols modes in HAM-lite:
Prescribed mean, standard deviation and (for mixed modes) species mixing ratios
DD -> Dust, SS -> Seasalt
BB -> Biomass burning
AV -> Anthropogenic+Volcanic (mainly sulphates)

Beautiful plots!



Interesting results!



Potential invigoration of convection with enhanced aerosol loading

What next?

Deeper process-level understanding of aerosol-cloud interactions, constraining model uncertainties with observations, influence of aerosol loading across different cloud types and environmental conditions, quantification of global ACRI

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